

The three vertebrate IP₃ receptors arose in two separate early vertebrate duplications

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Abstract

Vertebrates carry three inositol 1,4,5-trisphosphate receptor genes, *ITPR1*, *ITPR2* and *ITPR3*. Which two are sisters, when the duplications that made them happened, and whether they are products of the early vertebrate genome duplications have not been measured against a declared sample. We answered each question with a different instrument, and each instrument was run beside a control that could have failed. A maximum-likelihood tree of 134 representative proteins, rooted on the ryanodine receptors, makes *ITPR2* and *ITPR3* sisters. The approximately unbiased test rejects both alternative pairings ($p < 2 \times 10^{-5}$) and does not reject this one ($p = 0.476$). Reconciliation against a literature-calibrated species tree places the split of *ITPR1* from the other two on the vertebrate stem and the split of *ITPR2* from *ITPR3* on the gnathostome stem, bracketed 462–563 Ma, and the placement holds across five topologies and all 112 rootings. The genomic neighbourhoods are paralogous: across 309 genomes the *ITPR1* neighbourhood carries a paralogue of the *ITPR2* neighbourhood in 141 of 175 genomes and of the *ITPR3* neighbourhood in 89 of 152, against 2.6 % of matched random windows. *ITPR2* against *ITPR3* sits at that background, which does not corroborate the tree's sister pair. The teleost duplication doubled *ITPR1* alone, into two copies that partition one ancestral block. The three paralogues share a median 46 to 49 intron positions where 0.55 are expected, in every genome tested. The ryanodine receptors, which carry every Pfam domain the family is diagnosed by, share a median of one.

Introduction

Inositol 1,4,5-trisphosphate releases calcium from the endoplasmic reticulum [1] through a tetrameric channel of about 2,700 residues per subunit [2, 3]. From its first sequence the channel was recognised as the smaller relative of the ryanodine receptor [4], and the two families share the MIR, RIH and RIH-associated domains and the six-transmembrane pore. In vertebrates the IP₃ receptor is encoded by three genes, *ITPR1*, *ITPR2* and *ITPR3* [5], which differ in tissue distribution [6, 7], in single-channel behaviour [8] and in the diseases their variants cause [9, 10, 11].

How one ancestral gene became three is usually answered by assumption. The vertebrate lineage went through two rounds of whole-genome duplication [12, 13], and the teleost fishes through a third [14, 15], so a three-member vertebrate family with a single invertebrate orthologue reads naturally as the product of those events. Three questions are hidden in that reading, and none of them has been measured for this family. The first

is the branching order: which two of the three paralogues are sisters, which no published, support-annotated maximum-likelihood analysis rooted on the ryanodine receptors has fixed. The second is the timing: where on the vertebrate tree each duplication falls, which a gene tree cannot say on its own. The third is the mechanism: whether the duplications copied the surrounding genome as well as the gene, which is the evidence that separates a whole-genome duplication from a local one.

Each question needs a different kind of evidence, and each kind can mislead in a way the others cannot detect. A sequence tree can be pulled by long branches, by model violation and by mislabelled tips. A reconciliation inherits every error of the gene tree and of the species tree it is mapped onto. A neighbourhood comparison can be driven by annotation depth or by naming conventions rather than by the genome. We therefore used three instruments that share as little as possible: a tree, a dated reconciliation, and the genomic neighbourhood read in two independent ways. A fourth character, the positions of the introns, is one no protein alignment produces. Every one of them was run beside a control that could have failed.

The control that matters most is built into the family. The ryanodine receptors carry every Pfam domain used to diagnose the IP₃ receptors, which makes them the hardest decoy for any search, as the companion paper on the family's range shows (companion paper 1). Here the same property makes them the most useful control. They root the tree, they are a second three-member vertebrate family whose origin in the early duplications is not in doubt, and they pass through every instrument in the same genomes as the IP₃ receptors. Where an instrument reads the ryanodine trio correctly, its reading of the IP₃ trio can be taken at face value. Where it cannot, the limit belongs to the instrument.

The scope is declared and finite. The tree is built on 134 representative proteins chosen by stated rules from a census of 18,065 records. The neighbourhoods are read at every IP₃ and ryanodine receptor locus recovered from 309 vertebrate genome assemblies, and the gene architecture in the 189 of those assemblies contiguous enough to carry the whole gene. The species tree is an input taken from the literature and is not estimated here.

Results

A tree rooted on the ryanodine receptors makes *ITPR2* and *ITPR3* sisters

The alignment is built to root the tree, not to find the family. We chose 134 representatives from the census by eight coded rules, one per clade and paralogue cell, with the non-vertebrate grades and six ryanodine receptors included. No rule ranks on length or identity, because a longest-first pick favours chimeric gene models. MAFFT L-INS-i gave 11,777 columns at 76.23 % gaps, and trimAl kept 1,797 of them (15.3 %), 96.8 % of which are parsimony-informative (Extended Data Fig. 1; Supplementary Fig. 1). The family separation is measured again on this alignment, as a check that the outgroup is where it should be: mean identity is 0.910 within a vertebrate paralogue group and 0.253 from each group to the ryanodine receptors (Extended Data Fig. 2).

All three paralogues and the outgroup are clades. Maximum likelihood under Q.insect+R7 recovers the ryanodine receptors as a clade at the root and *ITPR1*, *ITPR2* and *ITPR3* each as a clade inside the vertebrates (Fig. 1). Each clade also takes in shark, chimaera and coelacanth genes whose records name no paralogue: seven in all, placed by the tree rather than by a name. With them, all three clades are supported at 100/100, and only six vertebrate genes, all from hagfish and lamprey, fall outside every clade. Of the 131 internal nodes, 91 (69.5 %) clear both SH-aLRT $\geq$ 80 and UFBoot $\geq$ 95, the joint bar every claim below is held to (Extended Data Fig. 3a). The tree overturns no census name: 39 vertebrate tips agree with their label, 0 are reassigned, and 5 are left unplaced. For each of the five, a reciprocal best-hit search against the human proteome upholds the database name, so the refusal to place reflects the tree's uncertainty rather than a mislabelled record (Extended Data Fig. 3b).

The sister pair is *ITPR2* with *ITPR3*, and the other two pairings are rejected. Each of the three rooted pairings was realised as a constrained maximum-likelihood search in which all three paralogue clades are mono-

phyletic, so the test compares the sister arrangement and nothing else. Under the approximately unbiased test over 10,000 RELL replicates, *ITPR1*+*ITPR2* costs 114.02 log-likelihood units and is rejected ($p = 1.8 \times 10^{-5}$), *ITPR1*+*ITPR3* costs 113.98 and is rejected ($p = 1.65 \times 10^{-5}$), and *ITPR2*+*ITPR3* is not rejected ($p = 0.476$), nor is the unconstrained tree ($p = 0.525$) (Fig. 2). The 95 % confidence set of topologies therefore holds 2 trees, and both carry the same pair. Mean identity had pointed the other way: *ITPR1* and *ITPR2* are the most similar pair on the alignment (Extended Data Fig. 4). That ranking ignores the outgroup and rate variation, which is why it was treated as a preview and the tree as the answer.

The answer does not depend on the model. Ultrafast bootstrap is optimistic under model violation, and an alignment that spans four kingdoms violates any single model. We re-ran the search with an extra nearest-neighbour-interchange round on every bootstrap tree and asked the same clade questions of the result, with clade membership read from a committed table so that bookkeeping cannot masquerade as topology. Nine of the claim clades are held and none is weakened or lost. The one clade that is poorly supported in the main tree, *ITPR1*'s census-labelled core, is poorly supported in both, while the same clade with the unlabelled tips included is at maximal support.

The cyclostome loci form two ancient lineages of their own. Hagfish and lamprey each carry three IP_3 receptor loci, and all six sit in cyclostome-only clades rather than inside any gnathostome paralogue clade. There are two such clades, each holding both species, so the cyclostome copies are two lineages that separated before hagfish and lamprey did, not an expansion either genome made on its own.

The three neighbourhoods are paralogous, and the paralogy runs through *ITPR1*

Each paralogue has a neighbourhood of its own. We read the flanking genes around all 2,144 IP_3 and ryanodine receptor loci in the 274 genomes that carry an annotation, and scored every pair of loci against matched random windows in the same two genomes, which holds annotation depth, naming convention and window rule constant. Within-paralogue similarity runs 216- to 411-fold above its own null, and every cross-paralogue and cross-family class sits at or below the null; over 168,241 pairs of IP_3 and ryanodine receptor loci the largest class mean is 0.0002 (Extended Data Fig. 5a). A paralogue caller built on these neighbourhoods alone, with its operating point set by maximising call rate minus false-call rate, is correct on 405 of 405 calls over 503 loci whose paralogue the assembly's own annotation establishes, and calls 6 of 726 random windows (Extended Data Fig. 5b). It then places 131 loci that the sequence sweep could not label. The same instrument shows that the *ITPR3* neighbourhood, which lies in the major histocompatibility region in human, is the one that decays fastest across vertebrate classes (Extended Data Fig. 5c).

Two families of flanking genes survive between the three neighbourhoods, and both touch *ITPR1*. Shared gene names are no test of whole-genome duplication, because after about 500 million years the flanking copies are paralogues of one another under different names. At the level of the gene family, with the trailing digits of each symbol removed, two links survive across the vertebrates: a metabotropic glutamate receptor family between *ITPR1* and *ITPR3*, and a basic helix-loop-helix family between *ITPR1* and *ITPR2* (Fig. 3). *ITPR2* and *ITPR3* share nothing at any threshold, and neither does any IP_3 and ryanodine receptor pair.

The paralogy replicates across 309 genomes against a real-window null. To test paralogy rather than shared names, we took the human paralogy map from a pinned Ensembl Compara release, removed every IP_3 and ryanodine receptor gene from every window so that the family's own paralogy could not count as evidence, and dated each link by its Compara duplication node. Asked of the flank sets in every swept genome, the *ITPR1* neighbourhood carries a paralogue of the *ITPR2* neighbourhood in 141 of 175 genomes (80.6 %, ten vertebrate classes) and of the *ITPR3* neighbourhood in 89 of 152 (58.6 %, nine classes). A null of 932 matched random neighbourhoods drawn by the same sampler in the same genomes carries a link 2.6 % of the time. *ITPR2* against *ITPR3* is 4 of 149 (2.7 %), which is the background ($p = 0.554$) (Fig. 4a).

Dating the links removes one of them from the two-round account. The glutamate receptor link between *ITPR1* and *ITPR3* is dated to Vertebrata, which is what an ohnologue pair from the early duplications looks like, and it is carried in 84 genomes. The helix-loop-helix link between *ITPR1* and *ITPR2* is dated to Opisthokonta: it is real paralogy, far older than the vertebrates, whose copies happen to lie beside these genes, and it is dated to Vertebrata in only 2 genomes. The ryanodine receptors pass through the identical instrument in the same genomes as the positive control, and in the single human genome they fare no better: at the primary window each family keeps one vertebrate-dated link between two of its three neighbourhoods, and neither survives correction across every test the stage ran (Fig. 4b). What the human genome alone measures is the instrument's ceiling, which is why the replication carries the claim.

The neighbourhood does not corroborate the tree's sister pair. The tree makes *ITPR2* and *ITPR3* sisters, and they are the one pair whose neighbourhoods keep nothing above background. The two instruments do not measure the same quantity. A tree estimates the order in which the copies diverged. A retained flanking ohnologue records which copies survived deletion beside each gene, and the quartets left by whole-genome duplication lose their flanking copies lineage by lineage with no memory of the duplication order [16, 17]. We therefore report the disagreement as a lack of corroboration and not as a contradiction.

Reconciliation places the two duplications on different branches

The species tree is an input with its uncertainty attached. We mapped the gene tree onto a hand-curated species tree for the 31 sampled vertebrate species, with 29 named internal nodes, each carrying a published crown age, the spread of published estimates, a stem age and its source [18, 19, 20]. Duplications were called by the non-binary rule [21], which is required because the familiar binary rule reads a support-collapsed polytomy as a speciation, and losses were counted by the standard method [22, 23].

The earlier split is on the vertebrate stem, the later one on the gnathostome stem. On the reported tree with every tip included, the split separating *ITPR1* from *ITPR2* and *ITPR3* maps to the vertebrate stem, older than crown Vertebrata. Published estimates for that crown run from 480 to 615 Ma around a point age of 563 Ma [24, 25], and nothing in the sampled tree bounds the duplication from above. The split separating *ITPR2* from *ITPR3* maps to the gnathostome stem, between crown Gnathostomata at 462 Ma and crown Vertebrata at 563 Ma (Fig. 5). The two duplications sit on different branches, and that makes *ITPR1* the earlier-diverging copy.

The placement survives every perturbation we could make to the gene tree. We reconciled five topologies (the reported tree, the three constrained sister hypotheses and the model-violation re-search) under three tip treatments, twelve reconciliations in all. With the cyclostome loci kept, the deepest paralogue duplication maps to Vertebrata in all 5 topologies, including the two sister arrangements the AU test rejects, so the dating does not depend on which pair is sister (Extended Data Fig. 6a). Collapsing every node below the support bar keeps the placement, even though the gene-tree node that unites *ITPR2* and *ITPR3* with *ITPR1* is the weakest in the vertebrate subtree (17.4/54). Re-rooting the vertebrate subtree at every one of its 112 edges leaves the placement where it is: the minimum-event rooting needs 64 events against 67 for the outgroup rooting, and both put the deepest duplication at Vertebrata (Extended Data Fig. 6a).

What forces the older placement is where one cyclostome lineage attaches. One of the two cyclostome clades sits among the gnathostome paralogues, sister to *ITPR2* and *ITPR3* at maximal support. A lineage holding both cyclostome and gnathostome copies forces every duplication above it to predate the split between the two groups. The same attachment keeps the later duplication on the gnathostome stem, because the node uniting *ITPR2* and *ITPR3* then holds gnathostomes only. Dropping the six cyclostome tips moves the deepest duplication to Gnathostomata, so the result rests on six tips and we measured the standard objection to them. Cyclostomes are the classic long-branch risk, and long branches are drawn to the root, where this answer sits. Their root-to-tip distances lie at 0.957 to 1.045 times the median of all 57 vertebrate tips, and none is an outlier (Extended Data Fig. 6b).

The implied losses count sampling, not biology. The reconciliation implies 47 lost cells. Checked one at a time against the genome sweep, 26 are the paralogue present in the genome and absent only from the 134-tip sample, and 0 are corroborated (Extended Data Fig. 6c). We report no losses from it; a loss count needs the genomes themselves.

The teleost duplication doubled *ITPR1* alone, in one ancestral event

Five predictions of the third genome duplication hold. Above the contiguity bar, every non-teleost gnathostome genome carries one copy of each paralogue and three ryanodine receptors. The four pre-duplication ray-finned genomes (bichir, gar and bowfin) carry one of each and three ryanodine receptors. The 73 teleost genomes carry a mean of 1.973 copies of *ITPR1*, 1.041 of *ITPR2* and 1.041 of *ITPR3*, with 97.3 % carrying two copies of *ITPR1*, while the same genomes carry 5.822 ryanodine receptors (Extended Data Fig. 7a). The lineages with a further duplication, sturgeon and salmon, carry three, two and two copies and eight ryanodine receptors [26]. The ryanodine count is what makes the singletons interpretable: the duplication copied *ITPR2* and *ITPR3* as surely as *ITPR1*, and the sister family kept its copies in the same genomes, so the missing second copies are a fact about retention, not about the search.

The two copies are dispersed, not tandem. Of 71 teleost genomes carrying two copies, 20 place them on different contigs, and the rest a median 8,870,578 bp apart, the closest pair anywhere at 535,374 bp.

The two copies partition one ancestral block. In 49 two-copy genomes with both copies flanked, the two copies' ancestral flanking symbols are disjoint in 45 against a tetrapod reference and in 46 against a pre-duplication ray-finned reference (Fig. 6). Disjoint is the load-bearing word. Two copies that each resemble the ancestor could be two later duplications; two copies that divide the ancestral neighbourhood between them are what reciprocal loss after a single duplication produces.

One duplication made both copies. Each genome's copy labels are arbitrary, so we let anchor genomes from 6 orders assign every genome's two copies independently and asked whether the anchors agree. Under independent lineage-specific duplications they would agree half the time, and a constructed control requires the statistic to land there. All 705 of 705 assignments agree ($p = 1.2 \times 10^{-212}$) (Extended Data Fig. 7b). A sequence call made with no neighbourhood input agrees too: in the six genomes whose two copies won different baits, the bait and the block agree six times out of six.

The three paralogues share one intron set, and the ryanodine receptors share none of it

The exon structure is read from the genome and checked against it. We measured the exon structure of every locus in the 189 assemblies contiguous enough to carry the whole gene, 1,378 genes and 112,254 junctions, from the same spliced alignments the sweep produced. What counts as an intron was set by the genome rather than chosen: every block gap was scored by its splice dinucleotides, and since no population of short unspliceable gaps exists, the floor went at the smallest gap the genome calls spliceable. 99.89 % of junctions read as a spliceable pair, with 1,035 at minor splice sites, and 177,710 of 188,146 coding edges annotated by independent pipelines in 164 genomes (94.5 %) fall exactly on a boundary the alignment placed (Extended Data Fig. 8b).

The exon count is conserved and the genomic span is not. The three paralogues carry a median 58, 57 and 58 coding exons around 8,250, 8,100 and 8,008 bp of coding sequence, at mean exon lengths of 141.4, 142.2 and 137.8 bp. Their median genomic spans are 147,445, 243,700 and 58,175 bp, a 4.19-fold range at a 3 % difference in coding length (Extended Data Fig. 8a). The ordering holds inside single genomes: *ITPR3* is the shorter gene than *ITPR1* in 161 of 182 genomes and than *ITPR2* in 146 of 181. The ryanodine receptors, measured through the same instrument, are 104-exon genes around 14,910 bp of coding sequence, at a mean exon length of 144.5 bp.

The intron positions are shared. We placed every intron in one coordinate frame, the representative alignment's columns, reached through each bait's human reference. The frame was checked before use: all 14 residues measured on the IP₃-bound structure (the ten ligand contacts and the filter and gate residues) land in one column in all three paralogues. An intron's position is the column its upstream exon ends in together with its phase, since two introns in the same place in different frames are not one ancestral intron. Scored inside each genome against an exact null that places each intron independently and uniformly on the columns both loci resolve, *ITPR1* and *ITPR2* share a median 48 positions, *ITPR1* and *ITPR3* 46, and *ITPR2* and *ITPR3* 49, where 0.55 are expected. These are 87.6-, 85.6- and 88.4-fold enrichments, significant in every one of the 183, 182 and 181 genomes tested (Fig. 7). Each paralogue carries a core of 48 to 49 positions present in over 90 % of its genomes.

The sister family shares a fold and none of the introns. Measured through the same instrument in the same genomes, an IP₃ receptor and a ryanodine receptor share a median of one intron position, significant in 0 of 188, 0 of 184 and 0 of 183 genomes (Fig. 7b). The ryanodine receptors have a core of their own, 91 positions present in over 90 % of their genomes, so the instrument reads their structure well. The two families share four Pfam signatures and a pore, and whatever their shared domain architecture means, it was not inherited as a gene structure.

The same measurement shows where fragmentary annotations stop. The assemblies' own annotations hold 291 of these loci as split or fragmentary gene models. Of 1,448 internal termini of those models, 689 fall on an exon boundary the gene carries, 406 fall inside an exon and 353 inside an intron, and 228 of the 291 loci carry a terminus that no splice site explains (Extended Data Fig. 8c). Only 3 are broken entirely at real junctions. A duplication detector built on the same blocks, scored against the committed copy count it never sees, finds 342 of 343 duplicated cells and wrongly calls 21 of 893 single-copy ones.

Discussion

Three instruments that share almost no evidence give a consistent account of where the three vertebrate IP₃ receptors came from. The genomic neighbourhoods are paralogous, with a vertebrate-dated ohnologue pair that replicates across the vertebrates against a real-window null, so at least one of the duplications copied a block of genome and not a single gene. The dated reconciliation puts the two duplications on two different branches: the first, which separated *ITPR1* from the ancestor of *ITPR2* and *ITPR3*, on the vertebrate stem, and the second on the gnathostome stem, bracketed 462–563 Ma. Both positions are where the two rounds of early vertebrate duplication are usually placed [13, 27]. The intron positions add a character that no protein alignment can produce: the three paralogues carry one inherited exon structure, conserved to within a few introns across 189 genomes, and the ryanodine receptors carry an unrelated one.

One disagreement stays open, and we report it rather than resolve it. The tree makes *ITPR2* and *ITPR3* sisters and rejects both alternatives. The neighbourhood makes *ITPR1* the block that kept a flanking ohnologue with each of the other two, and *ITPR2* with *ITPR3* the one pair that retains nothing above background. These are different quantities. The order of branching is a statement about when copies diverged, while a retained flank records which copies of each neighbour survived hundreds of millions of years of reciprocal deletion, a process that reconstructions of the vertebrate ancestral genome show to be lineage-specific and uncoupled from duplication order [17, 16]. The reconciliation is consistent with both readings, because it makes *ITPR1* the earlier-diverging copy, the copy that could keep a shared flank with each of the others. That is suggestive rather than a test. Preferring whichever instrument agrees with a headline is how an analysis of this size talks itself into a wrong answer.

The dating rests on six tips, and we say so. The vertebrate-stem placement is forced by one cyclostome clade that sits among the gnathostome paralogues. We measured the two standard objections, weak support and long

branches, and neither applies: collapsing every unsupported node keeps the placement, and the cyclostome tips sit at the median root-to-tip distance. The placement is still decided by where one two-tip clade attaches, and a denser cyclostome sample is the experiment that would test it.

The teleost result is clean and its cause is not in this evidence. The third duplication copied all three paralogues, as the doubled ryanodine receptors in the same genomes show, and *ITPR1* alone kept its second copy in nearly every teleost. The two copies divide one ancestral block and were made by one event. Why *ITPR1* alone was retained is a question for expression and function, which this paper does not measure.

The limits are the instruments' own. Four stand out. First, which of the two early rounds made which split cannot be separated here: the dated link is dated to Vertebrata, which covers both rounds, and separating them needs the cyclostome side of the quartet, where the neighbourhood caller could not reach, because a genome in which fewer than a third of the flanking genes carry a name gives the caller nothing to compare. Second, the genome-wide block scan finds no block that looks like the missing fourth member of the quartet. With one dated ohnologue pair surviving between the blocks that do carry a gene, a block that lost the gene too has nothing left to be recognised by, so this is a limit of the evidence and not a claim that the fourth slot never existed. Third, the single human genome is underpowered for the two-round test for both families alike, which is why the replication across genomes carries the claim. Fourth, the AU test's confidence set holds two trees, and both carry the same sister pair; a different alignment could move deeper nodes but would have to move the pair out of a set that has already excluded both alternatives.

This paper makes no claim about losses or function. It does not count losses: the reconciliation's implied losses are sampling or cannot be assessed, none is corroborated, and a loss count needs the genomes themselves. It does not say what the paralogues do differently. It answers where they came from, and the answer is that the three IP₃ receptors of vertebrates are ohnologues from two duplications on two different branches of the early vertebrate tree, carrying one ancestral exon structure that their sister family does not share.

Methods

Scope: Four denominators are used, and each is declared here. The tree is built on 134 representative proteins chosen from a census of 18,065 IP₃ and ryanodine receptor records assembled from reference proteomes, InterPro and genome sweeps (the census and its family call are described in companion paper 1). The neighbourhood analyses use every IP₃ and ryanodine receptor locus recovered by a protein-to-genome sweep of 309 vertebrate genome assemblies, one per vertebrate order plus every species the protein record left in doubt: 2,144 loci, of which those in the 274 assemblies that carry a gene annotation have a readable neighbourhood. The reconciliation uses the 57 vertebrate tips of the tree and a species tree of 29 named internal nodes. The gene architecture uses the 1,378 genes in the 189 assemblies whose contigs are long enough to carry the whole gene.

Representatives: Eight coded rules choose the set: a vertebrate grid of clade band by paralogue with human forced in, the teleost co-orthologue pairs as a stress test, the deep vertebrate grade (cyclostomes, cartilaginous fish, coelacanth) entered unlabelled because a paralogue label is a vertebrate concept, non-vertebrate metazoa by phylum, non-metazoan eukaryotes gated on per-record contamination verdicts, copy-number expansions, gene models no database holds, and the ryanodine outgroup. The quality key is lexicographic, so the audit names the component that decided each pick, and each group's length target is its own measured median. Six selection rules have constructed negative controls that run on every build.

Alignment: MAFFT L-INS-i [28] was run single-threaded, because at an automatic thread count its refinement is not reproducible on this machine, and the SHA-256 of input and output are recorded. A ragged alignment is a hard failure, since a silent MAFFT failure degrades to a star alignment. trimAl -automated1 [29] chose the kept columns; the heuristic was fixed in advance rather than tuned against the resulting tree.

Tree: An exhaustive ModelFinder scan [30] was started and abandoned after it projected about 21 hours, and

replaced by a two-stage greedy scan (every matrix at fixed rate heterogeneity, then every rate model on the winning matrix) with both tables committed. Free-rate models were kept because they beat gamma models by 682 BIC units on this alignment. IQ-TREE 2 [31] ran with 1,000 ultrafast bootstrap [32] and 1,000 SH-aLRT [33] replicates, threads and seed pinned. A node is well supported only when it clears both SH-aLRT ≥ 80 and UFBoot ≥ 95 . Parologue membership was corrected by a coded rule that keeps a tip inside its group's largest pure clade, moves a tip only into a well-supported clade of at least three members of one other parologue, and otherwise leaves it unplaced; every tip the tree disputes was checked by reciprocal best hits against the human proteome.

Sister test: Each of the three rooted pairings was a constrained maximum-likelihood search under the same model, with the constraint naming only the three parologue cores and the outgroup. A constraint that names a tip pins it outside every group it declares, and a first version that named all 134 tips rejected every pairing, including the one the tree holds at maximal support; a constructed control now fails any constraint that names a free tip. The approximately unbiased test [34] used 10,000 RELL replicates, and the 95 % confidence set of topologies was read off it. As a model-violation guard the search was repeated with `--bnni`, and every clade claim was re-asked of that tree from committed tip sets.

Neighbourhoods: Flanking genes were read from each assembly's own annotation under two window rules ($(\pm)10$ coding genes, and the 10 nearest informative symbols each side, since human annotation names about 97 % of coding genes and sea lamprey about 27 %) and three symbol vocabularies, the third stripping trailing digits so that an ohnologue pair is visible at all. Loci were taken from the per-genome sweep output rather than a per-cell table, so that a teleost cell holding two genes contributes both. Every real locus pair was scored against matched random-window pairs drawn in the same two genomes, seeded per accession, and compared by a sign test with ties dropped and counted. Two empty neighbourhoods score zero, not one, so that two unannotated genomes cannot form a perfect match; this is one of 11 constructed controls run on every build. The parologue caller builds a consensus per parologue with the query's own species removed, and its threshold maximises call rate minus the random-window false-call rate; a call at or below the highest score any random window reached is reported as inside the null.

Dated paralogy: Human paralogy came from a pinned Ensembl BioMart archive (Ensembl Genes 116) [35, 36], with the archive's registry committed, so a re-run scores the same windows against the same gene trees. Every IP_3 and ryanodine receptor gene was removed from every window. The null for the human test is a permutation over real genomic windows at the matched gene count, 5,000 permutations at each of three window sizes, which keeps the gene-family clustering a shuffled gene set discards. Each link carries its Compara duplication node. The replication across 309 genomes applies the human map to the flank sets of every swept genome and compares against matched random neighbourhoods drawn by the neighbourhood analysis's own sampler, unchanged. The ryanodine receptors pass through every step as the positive control, since their origin in the early duplications is not in question.

Reconciliation: The species tree covers the 31 sampled vertebrate species. Every internal node carries a crown age, the spread of published estimates, a stem age and a source, and a validator fails on an uncalibrated node, a point age outside its own spread, an age inversion or a species mismatch. Where the literature does not resolve an arrangement the node is left a polytomy. The stem age matters because the sampled tree omits unsequenced lineages, so a node's parent here is usually older than the split that created it. Duplications were called by the non-binary least-common-ancestor rule [21] and losses counted after Zmasek and Eddy [23]. Five topologies were crossed with three tip treatments (all tips, cyclostome tips pruned, nodes below the support bar collapsed), and the collapse refuses a tree with no support labels, since a constrained search writes none. As a rooting control, the vertebrate subtree was re-rooted at each of its edges and scored by total events beside the outgroup rooting. Long-branch attraction was assessed by root-to-tip distance over all vertebrate tips. Every implied loss was checked against the genome sweep. Fourteen groups of constructed controls run before anything is written.

Teleost copies: A copy is a locus whose model covers at least half of its bait over at least 500 aligned residues at 0.4 identity or more, and counts were recomputed at seven coverage bars. Copy dispersion was measured on the genome. Double-conserved synteny was scored against a tetrapod and a pre-duplication ray-finned reference, because teleost symbols diverge from tetrapod ones. Block consistency was tested by letting anchor genomes from different orders assign every genome's two copies independently, against a two-sided binomial at one half; a constructed control builds independent lineage-specific duplications and requires the statistic to land at one half.

Gene architecture: Exon blocks were read from the sweep's spliced alignments in target order, with model keys disambiguated across the two assemblies processed in chunks and splice dinucleotides read strand-corrected. The intron floor was calibrated by binning block gaps and scoring them against the genome's splice dinucleotides. Intron positions were carried into the representative alignment's columns through a map from each bait to its cell's human reference; the map is refused unless all 14 structurally measured anchor residues land in one column in all three paralogues. The shared-intron null is exact: each intron of one locus is placed independently and uniformly on the columns both loci resolve, keeping its phase, so the shared count is Poisson-binomial and its tail needs no simulation. A seeded permutation without replacement is run beside it. Every cross-paralogue comparison is paired inside one genome, sign-tested with ties dropped, and Benjamini-Hochberg corrected across the family of tests. The ryanodine receptors pass through the same steps as the control. Twenty-one constructed controls run before anything is written, and six deliberate breakages of the rules were all caught.

Reproducibility: Every table is written by a script and every report is rendered from committed tables. Every load-bearing number in this paper is declared in a claims ledger with its source table and the operation that recovers it, and is re-verified on every build.

The maximum-likelihood tree of 134 family proteins recovers ITPR1, ITPR2 and ITPR3 as clades, rooted on the ryanodine receptors



10

The AU test rejects two of the three sister hypotheses for ITPR1/2/3

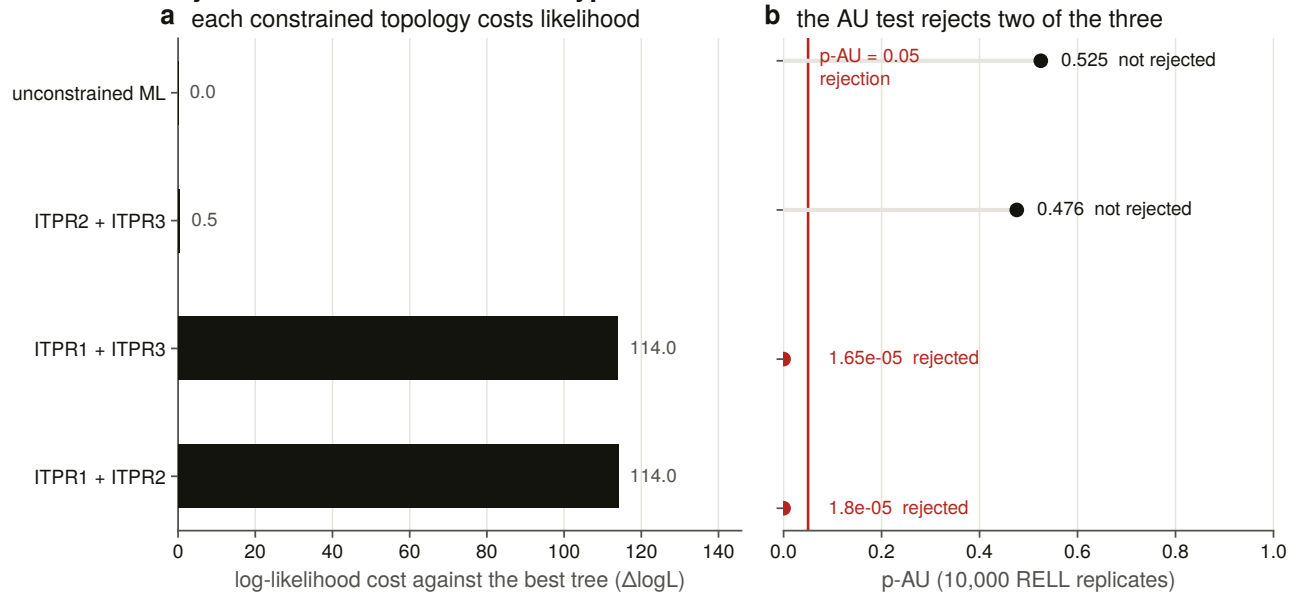
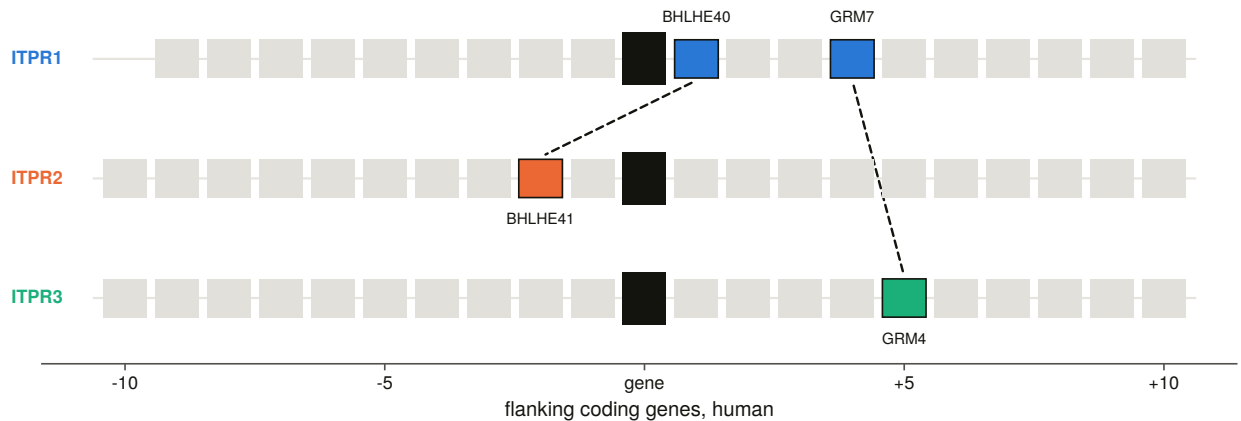


Fig. 2. Two of the three possible sister pairings fall outside the 95 % confidence set of topologies. (a) Each constrained topology is plotted by the log-likelihood it costs against the unconstrained tree. (b) The approximately unbiased p-value over 10,000 RELL replicates is plotted on the same row of categories, with the rejection line at $p = 0.05$. Two panels are used rather than twin axes because the two measures have unrelated scales. The pairing of *ITPR2* with *ITPR3* is the only one the test retains.

a The three human ITPR neighbourhoods share the shaded gene families with one another



b Each shared family (root key) is measured against its random-window background

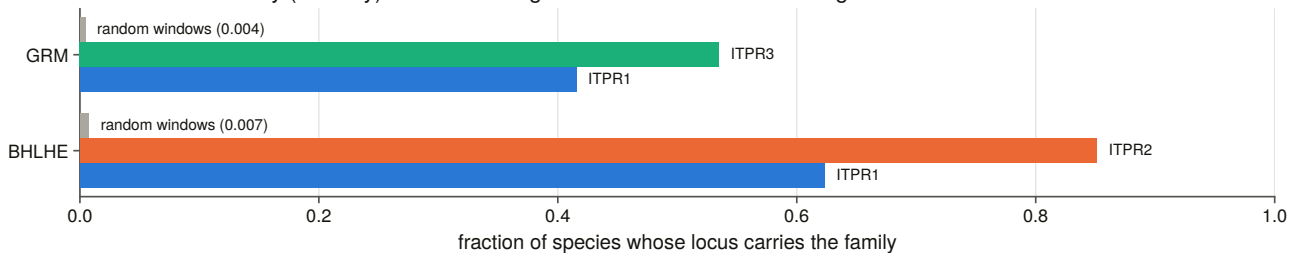


Fig. 3. Both flanking-gene families that survive between the three human neighbourhoods touch *ITPR1*. (a) The *ITPR1*, *ITPR2* and *ITPR3* neighbourhoods are drawn as gene tracks, with families shared between two loci shaded and linked. (b) Each shared family's prevalence across the swept genomes is set against the rate at which random windows carry it. The tracks are read from a committed table, so the figure cannot show a gene the data lack, and no shading connects the *ITPR2* and *ITPR3* tracks.

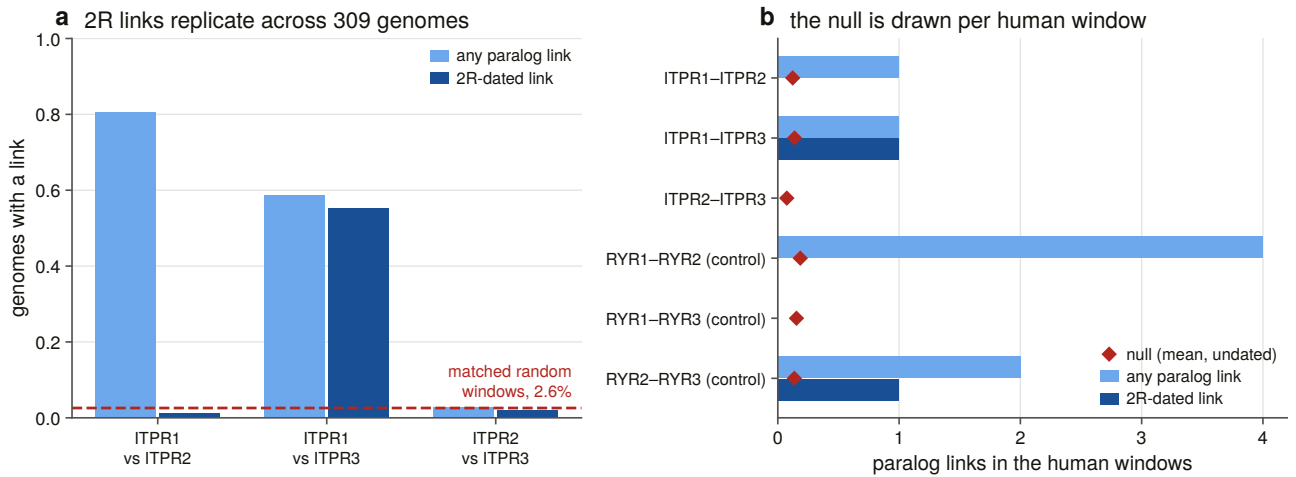


Fig. 4. The *ITPR1* neighbourhood is paralogous to both others across the vertebrates, while *ITPR2* against *ITPR3* sits at the background. (a) Bars give the fraction of genomes in which one neighbourhood carries a paralogue of the other, with the matched random-window rate drawn across them. (b) The single human genome is tested against a real-window permutation null, pair by pair, with the three ryanodine receptor pairs beside the IP_3 receptor pairs. The sister family behaves the same way through the same instrument, which is what makes the weak human result a property of the test.

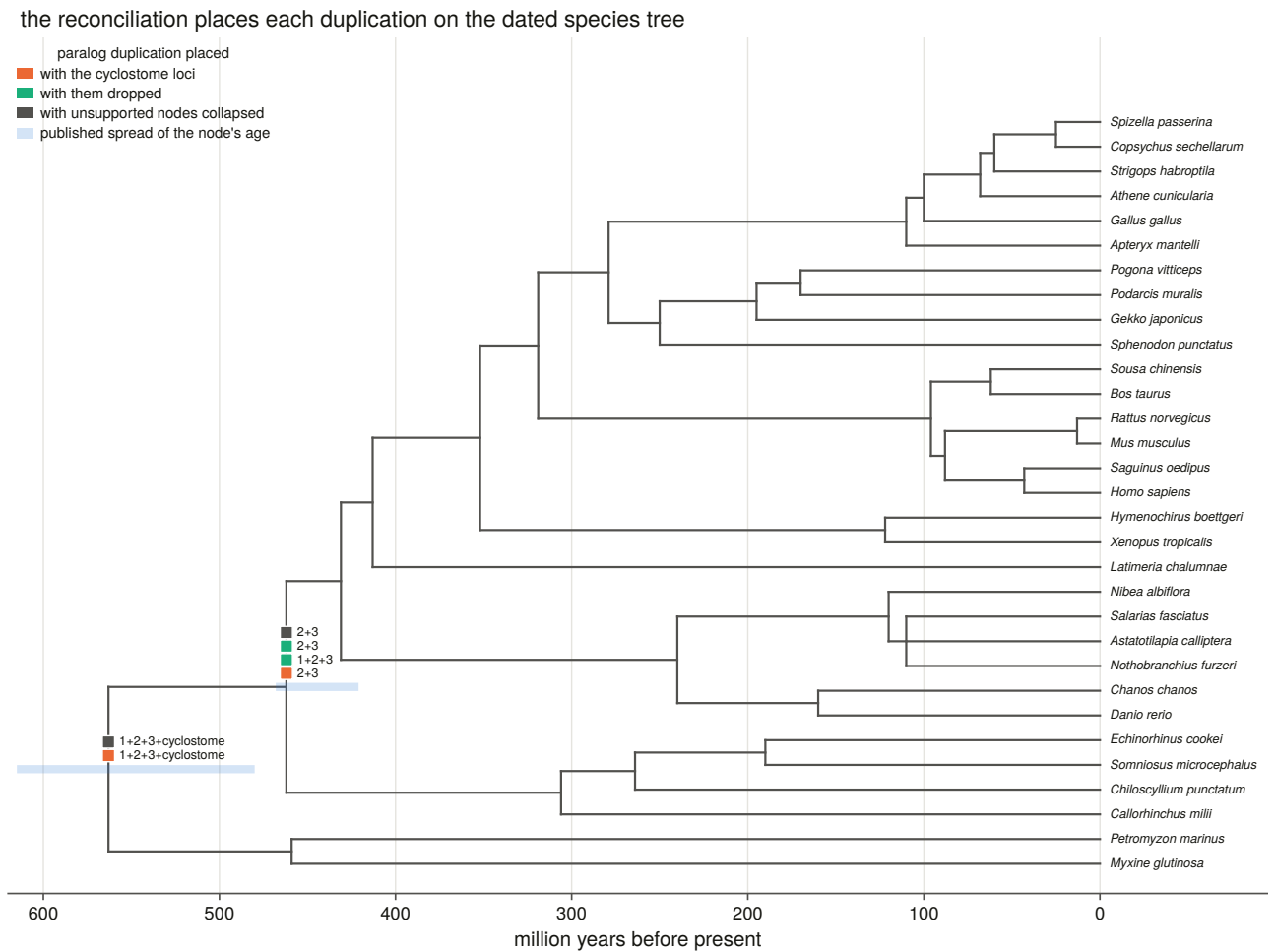


Fig. 5. The two duplications map to two different branches of the dated vertebrate tree. Nodes are placed on a linear time axis, and the spread of published ages is drawn as a band on the nodes a placement uses. Markers show each distinct placement once, with its tip treatment, since repeating the same placement for every matrix cell would make agreement look like weight. The older bracket is left open because nothing in the sampled tree closes it.

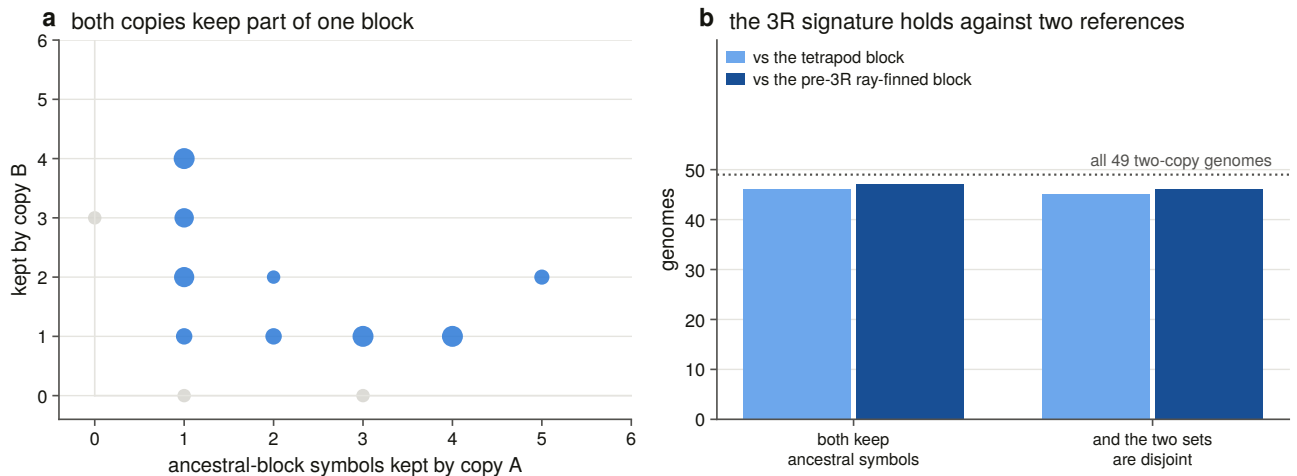


Fig. 6. The two teleost *ITPR1* copies divide one ancestral block between them. (a) Each two-copy genome is placed by how many ancestral-block symbols each copy retains, so a point away from both axes is a genome in which both copies keep part of the block. (b) Bars count the genomes meeting that condition and the stricter one that the two symbol sets are disjoint, against a tetrapod and a pre-duplication ray-finned reference, beside a line at all 49 genomes. Disjoint sets are what reciprocal loss after one duplication leaves.

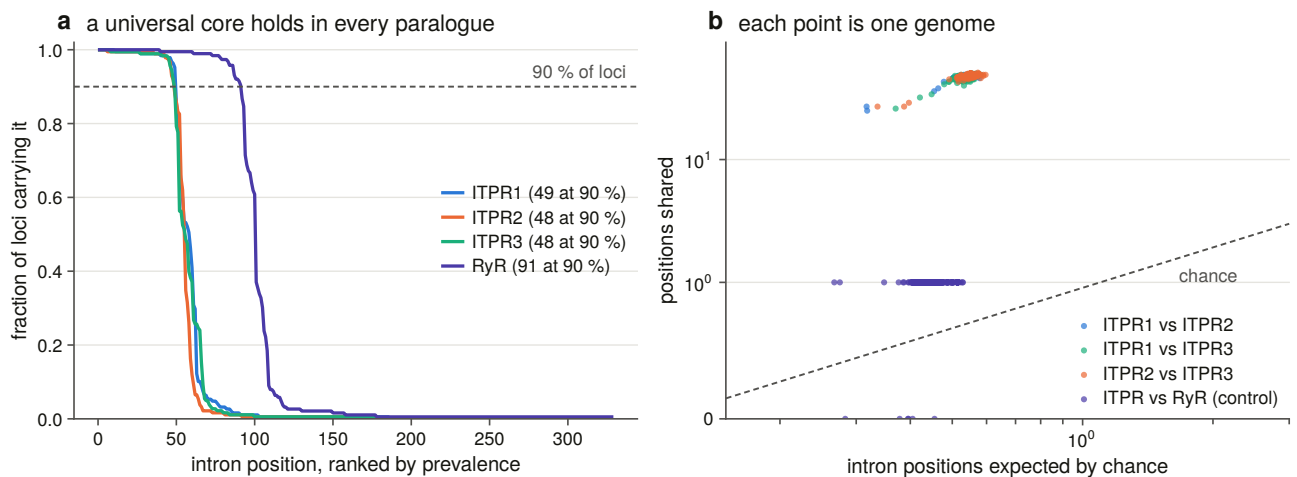
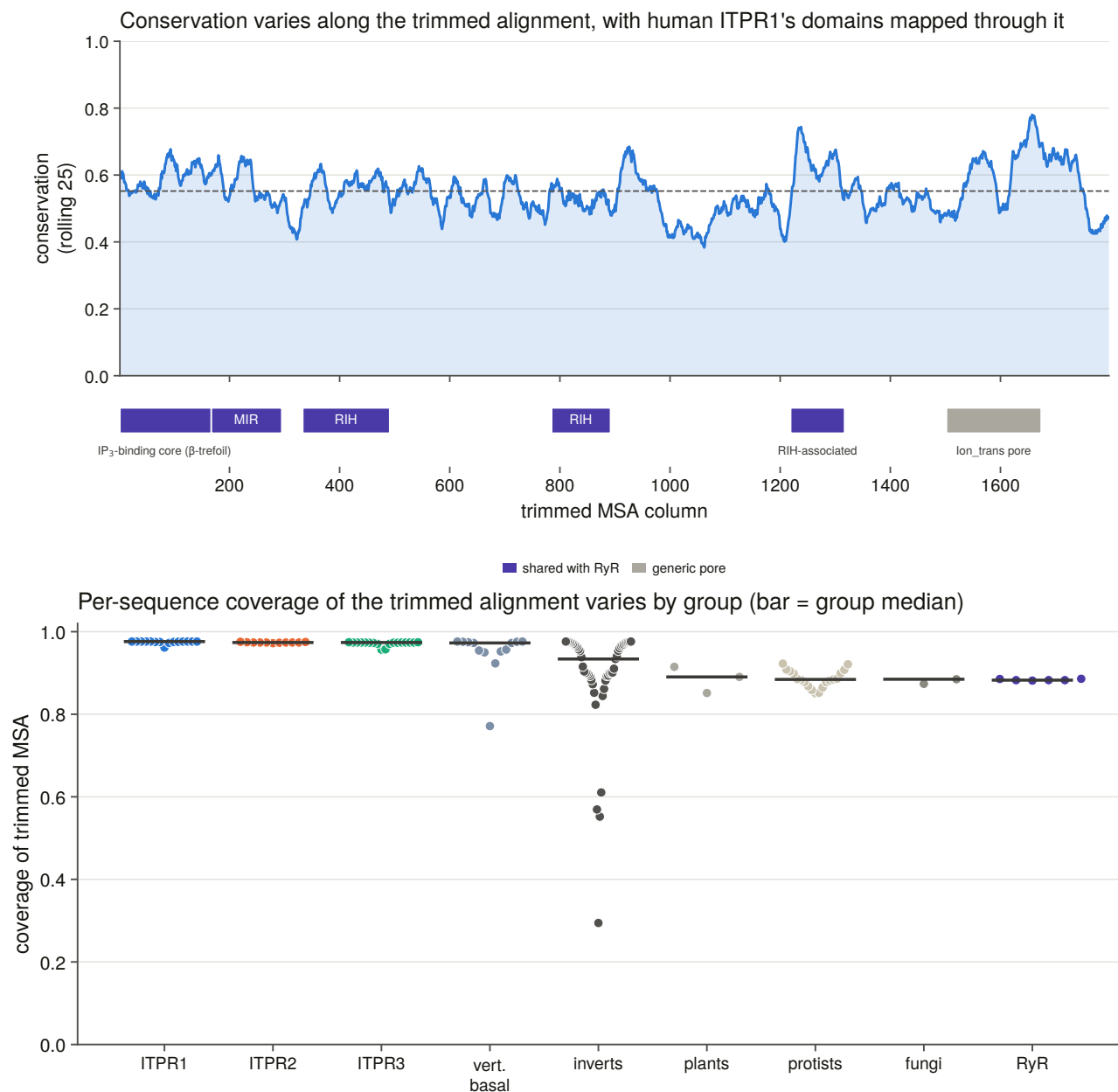


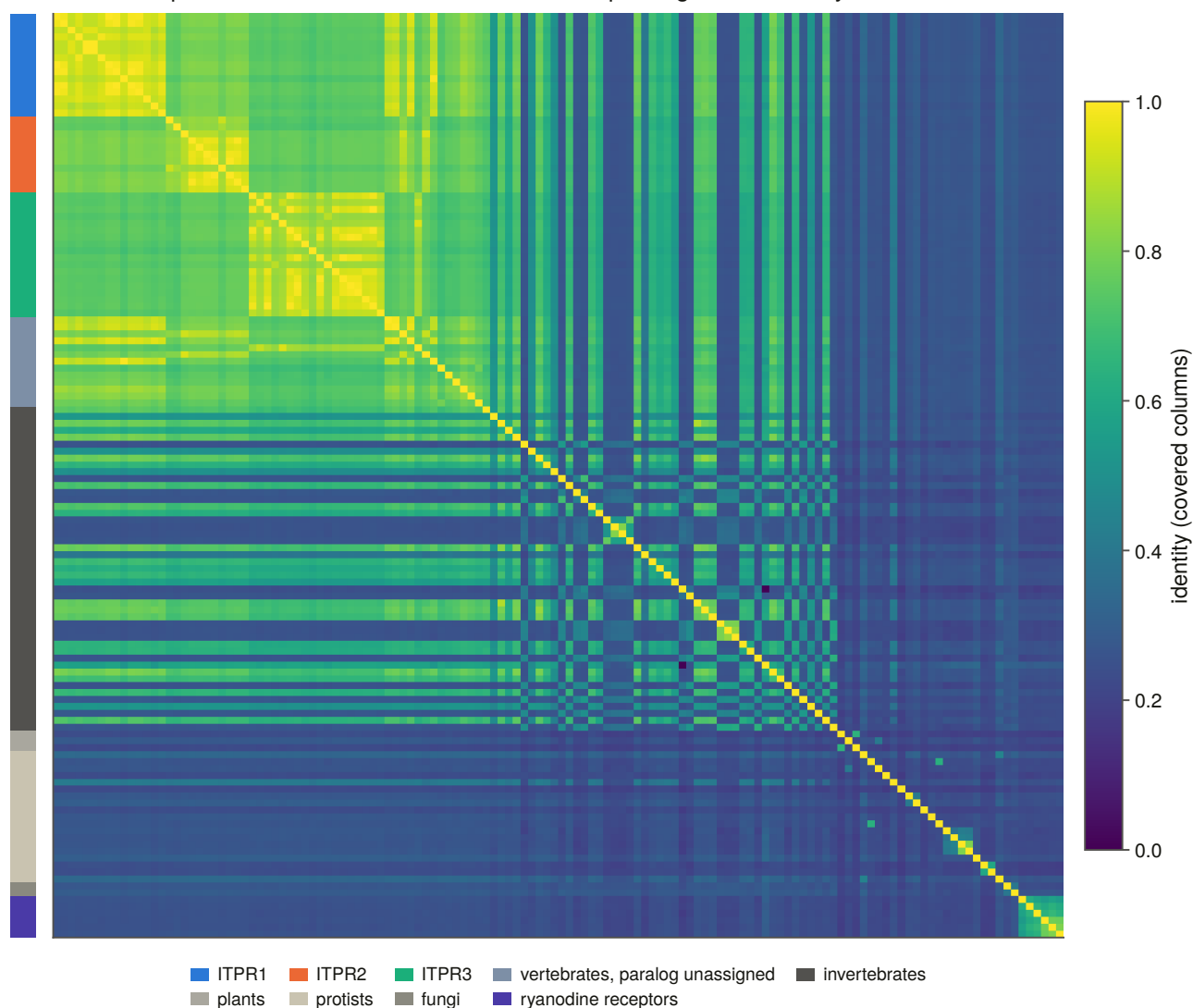
Fig. 7. The three paralogues share their intron positions in every genome tested, and the ryanodine receptors do not. (a) Intron positions within each paralogue are ranked by the fraction of genomes that carry them, with the 90 % line drawn. (b) Shared positions are plotted against the number expected by chance, one point per genome and pair, with the identity line. Every IP_3 receptor pair sits far above the line, and every pairing with the ryanodine receptors sits on it.

Extended Data

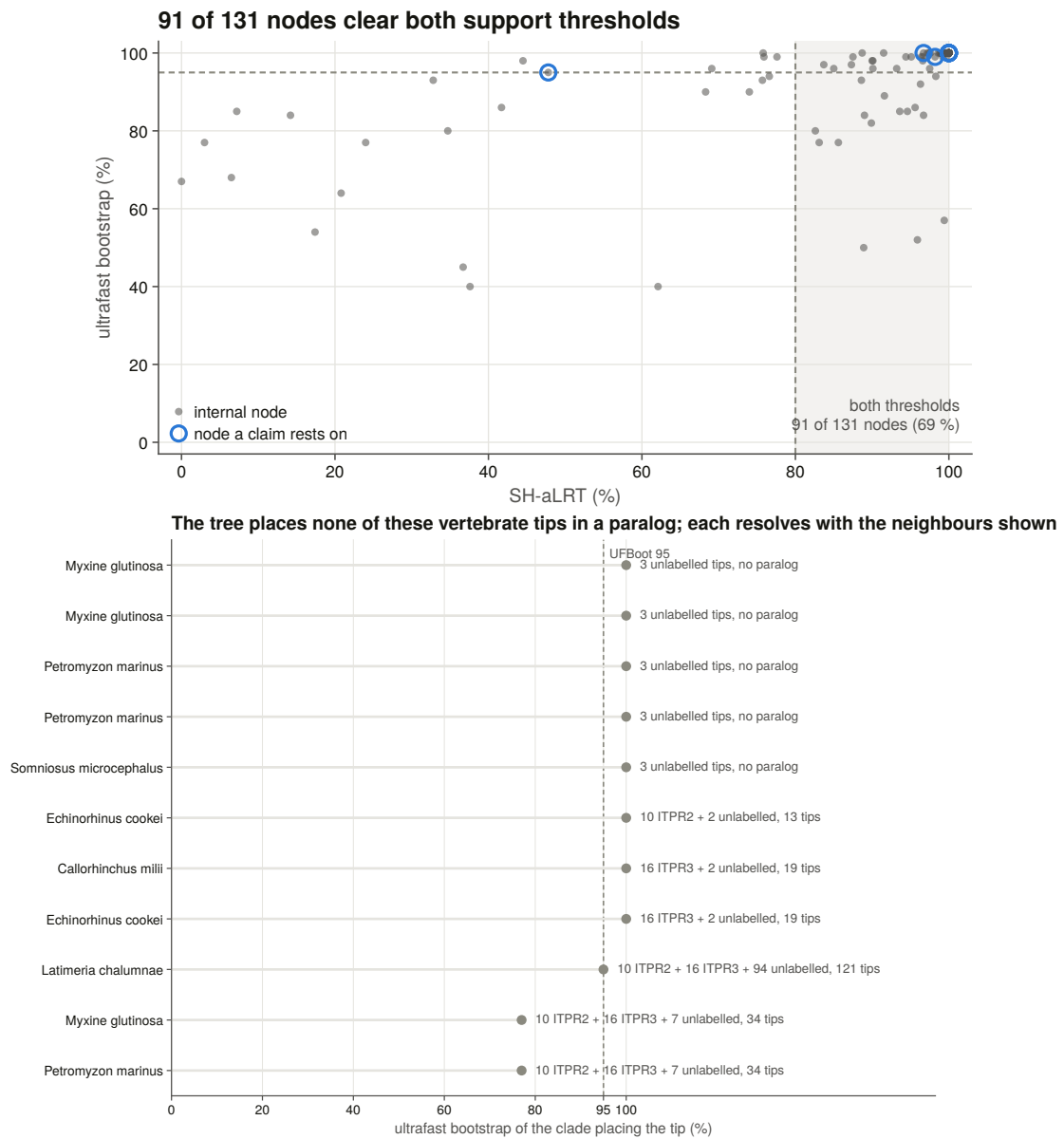


Extended Data Fig. 1. Trimming leaves an alignment that almost every representative covers. (a) Per-column conservation of the trimmed alignment is drawn under human *ITPR1*'s domain architecture, which is mapped through the alignment by walking the human row rather than by scaling residue coordinates onto columns. (b) Each representative's coverage of the trimmed alignment is plotted by group; the median is 0.96 and one tip of 134 covers less than half.

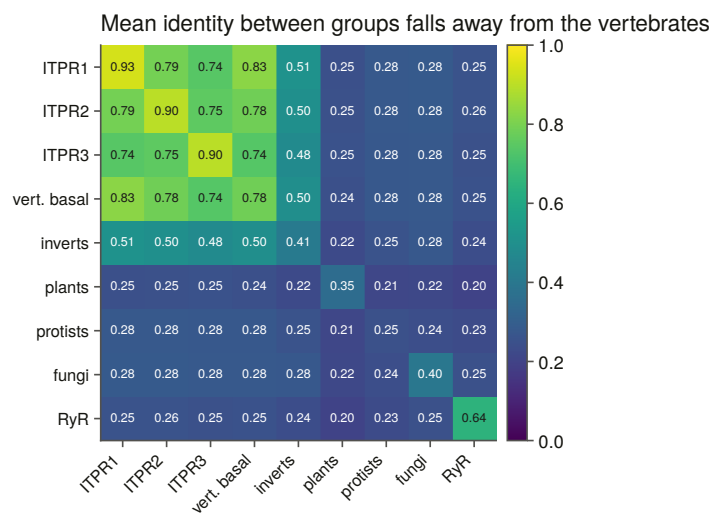
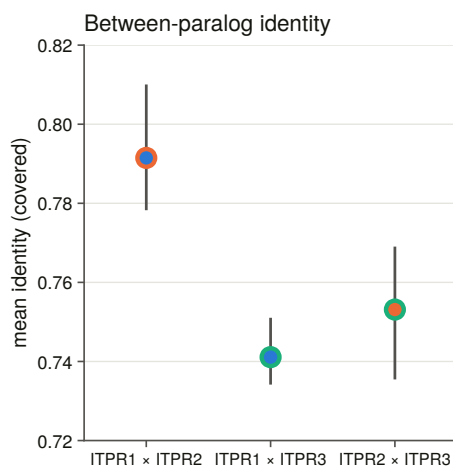
The 134 representatives are most identical within a paralog, over mutually covered columns



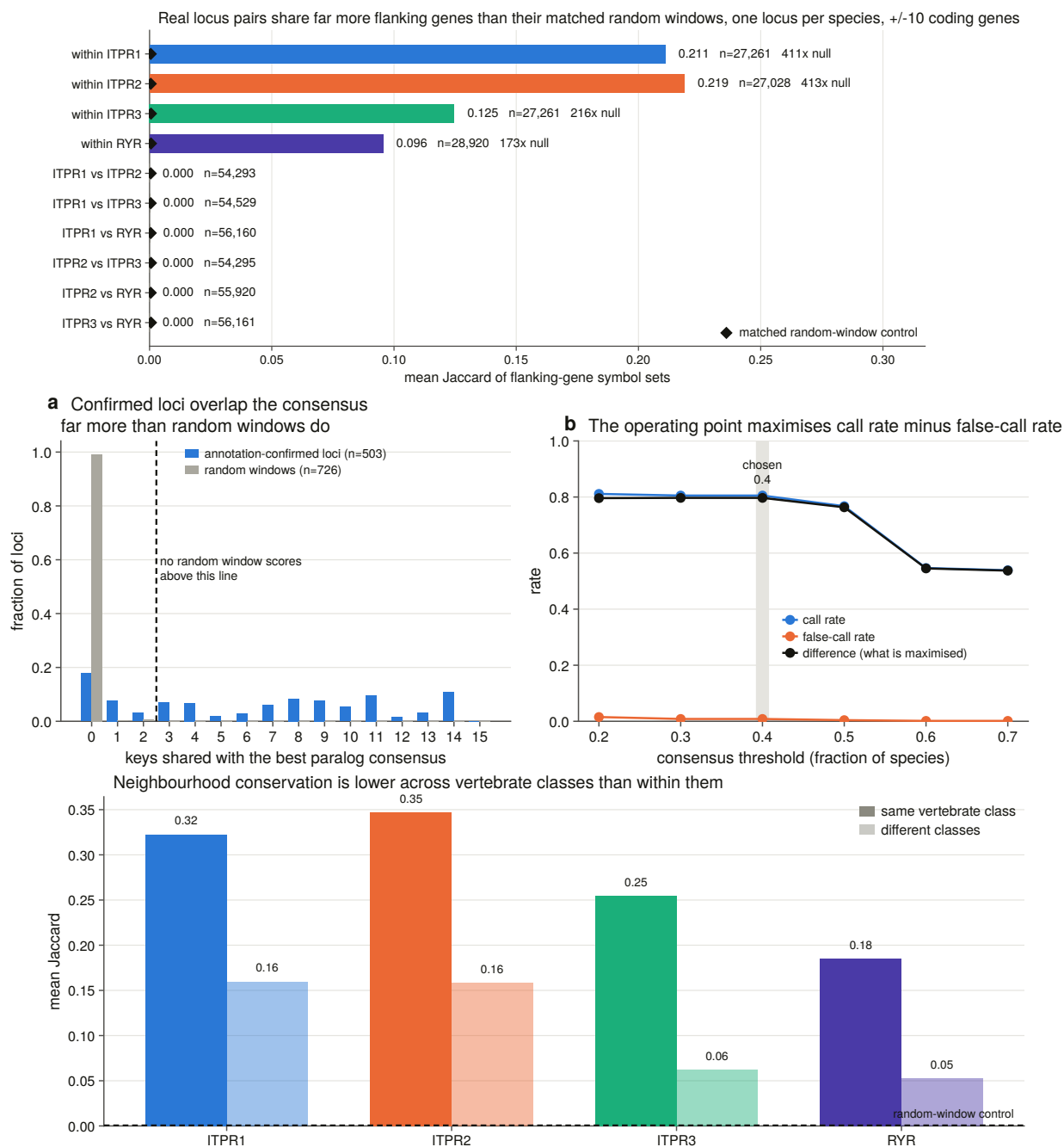
Extended Data Fig. 2. All-pairs identity over mutually covered columns separates the vertebrate paralogue groups from each other and from the outgroup. The matrix is ordered and side-barred by group, and the three vertebrate paralogue blocks stand out from a dark cross-family field. Gaps are not counted as mismatches, so a partial sequence does not look divergent.



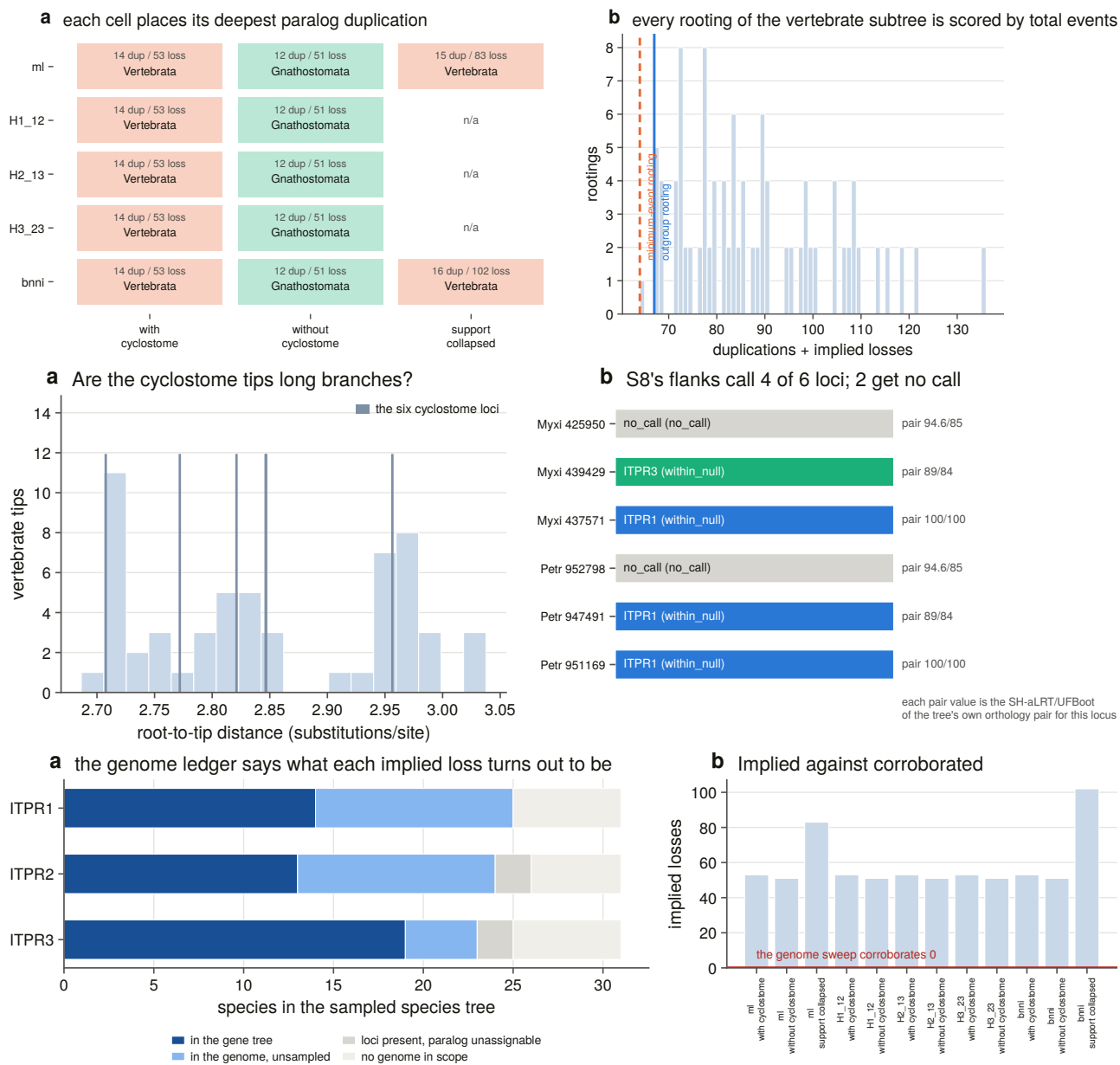
Extended Data Fig. 3. Most nodes clear both support tests, and the tree disputes no census name. **(a)** Every internal node is plotted by SH-aLRT against ultrafast bootstrap, with the joint threshold region shaded and the claim nodes ringed. **(b)** The vertebrate tips the tree places in no paralogue clade are listed with the support of the clade each lands in; reciprocal best hits uphold the database name for all five disputed names.



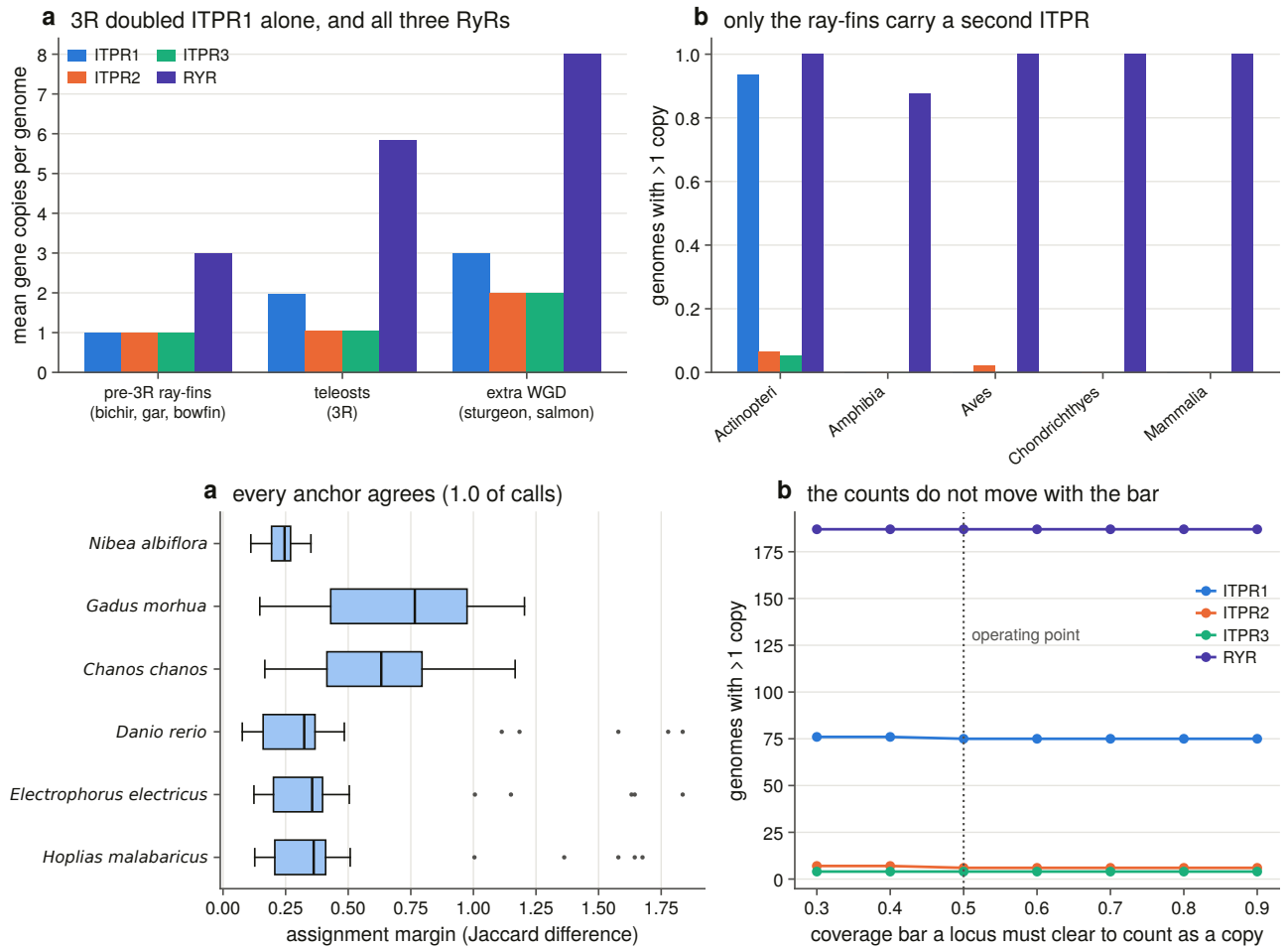
Extended Data Fig. 4. The three paralogues sit at 0.74 to 0.79 identity to one another and 0.91 within themselves. The left panel ranks the three paralogue pairs and the right panel puts every group pair on one scale. By identity alone *ITPR1* and *ITPR2* are the closest pair, which the tree contradicts.



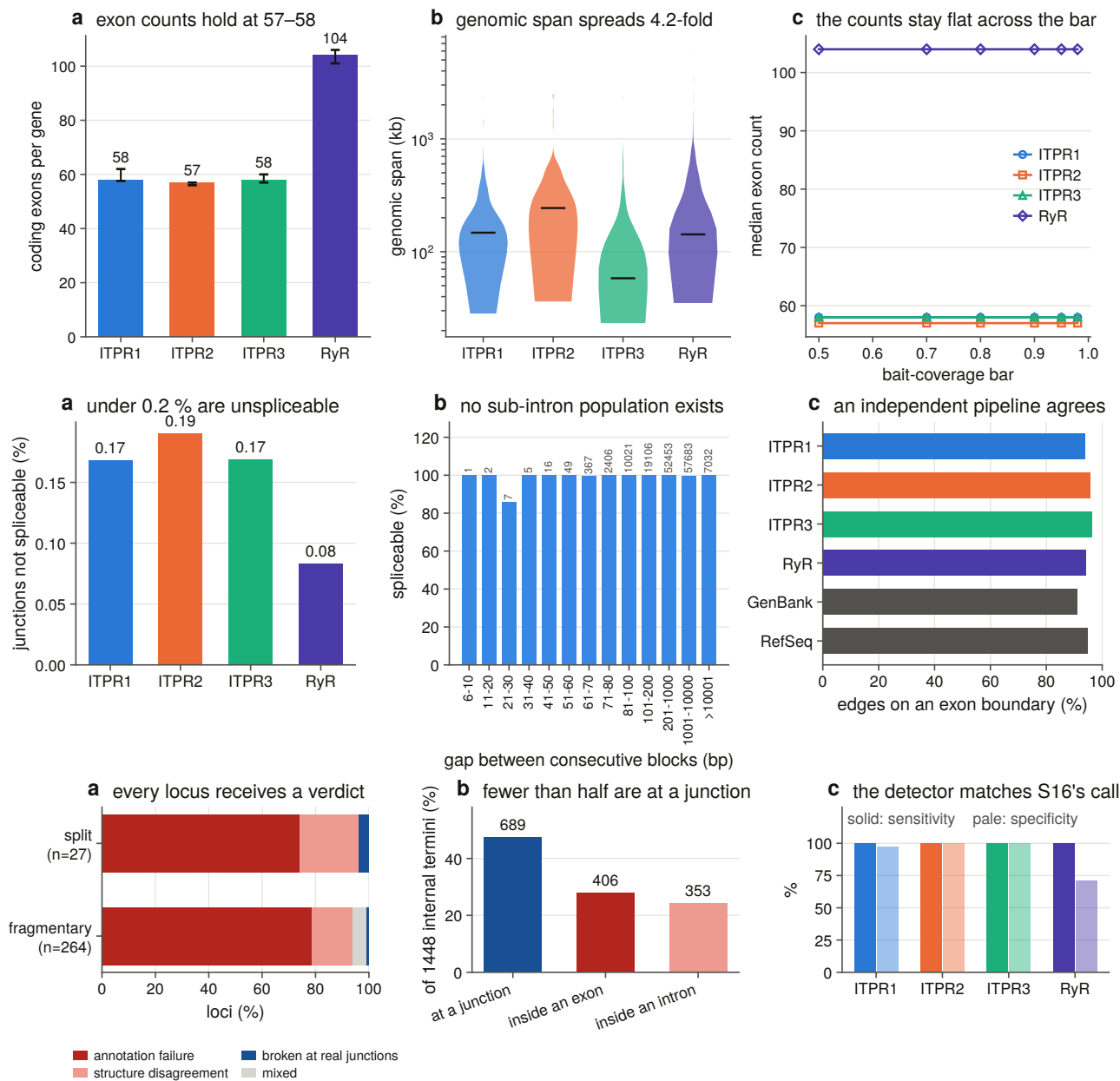
Extended Data Fig. 5. Neighbourhood similarity confirms the paralogue cells and finds nothing across the family boundary. **(a)** Mean flanking-gene similarity is shown for each pair class beside its matched random-window control, one locus per species. **(b)** The paralogue caller's scores on annotation-confirmed loci and on random windows are shown with the threshold sweep, where the operating point is set at the largest gap between call rate and false-call rate. **(c)** Within-paralogue similarity is split into pairs inside one vertebrate class and pairs across classes, where *ITPR3*'s neighbourhood falls well below its two siblings.



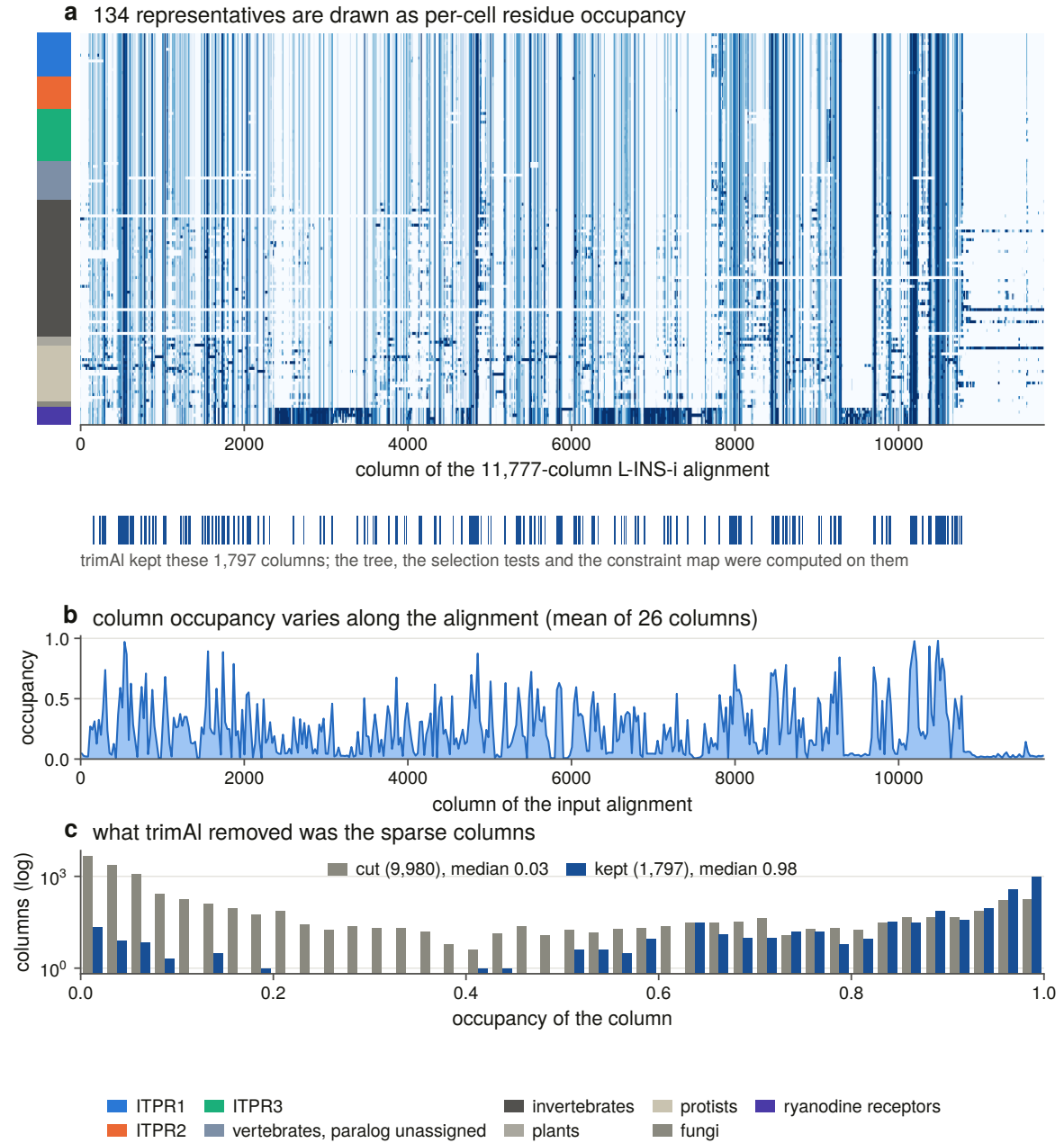
Extended Data Fig. 6. The dating holds across topologies, rootings and tip treatments, and a loss count read off it would count sampling. **(a)** Each cell of the five-topology by three-treatment matrix names its deepest paralogue duplication, and every rooting of the vertebrate subtree is scored by total events with the outgroup and minimum-event rootings marked. **(b)** Root-to-tip distances of all 57 vertebrate tips are drawn with the six cyclostome loci marked, beside the neighbourhood call for each locus. **(c)** The implied losses are classified against the genome sweep, and none is corroborated in any cell of the matrix.



Extended Data Fig. 7. Copy number follows each lineage's duplication history, and the teleost copies come from one event. **(a)** Mean copies per genome of each paralogue and of the ryanodine receptors are grouped by whole-genome duplication status, beside the fraction of genomes carrying more than one copy by vertebrate class. **(b)** Each anchor genome's cross-anchor assignment margin is positive, and the multi-copy counts stay flat across seven coverage bars.



Extended Data Fig. 8. The exon count is conserved while the genomic span is not, and the instrument that measures both is checked against the genome. **(a)** Coding exons, genomic span on a logarithmic axis, and the median exon count across six coverage bars are drawn for each paralogue and the ryanodine control. **(b)** The unspliceable fraction of junctions is drawn on the complement, by gap bin, and beside agreement with independently annotated coding edges. **(c)** Every split and fragmentary annotated model receives a verdict, its internal termini are placed against the exon structure, and the duplication detector is scored against the committed copy count.



Supplementary Fig. 1. Trimming kept the dense columns and cut the sparse ones. The input alignment's occupancy is rasterised in its own coordinates for all 134 representatives, its mean occupancy is plotted along the same axis, and the kept columns are ticked beneath. Only one raster is drawn because the trimmed alignment shares no x-axis with its input.

Data availability

Every table and figure this paper reports is committed under `results/` in the project repository, in the directories `msa_v2`, `phylogeny`, `synteny`, `duplication`, `reconciliation` and `gene_architecture`, with the inference limits in `methods`. The paper's deposit manifest lists every deposited file with its SHA-256. Genome assemblies, proteomes and the per-genome sweep output are not redistributed; they are retrieved from NCBI and UniProt by the accession lists deposited beside them.

Code availability

The scripts that produce every table are in `scripts/`, one prefix per analysis (`s6_`, `s7_`, `s8_`, `s13_`, `s16_`, `s21_`). Each analysis runs its constructed negative controls before it writes anything. `python scripts/s26_assemble.py --paper origin` rebuilds this paper from the committed tables and fails on a number that no longer matches its source.

Author contributions

George Dickinson directed the project, set its questions and reviewed its results. The analysis code and the text of this paper were written by Claude (Anthropic) under his direction; Chapter 16 of the accompanying thesis describes how the work was carried out.

Competing interests

The author declares no competing interests.

Companion papers

1. George Dickinson. The IP₃ receptor is ancestrally eukaryotic and has been lost repeatedly outside the animals. *In preparation* (paper 1 of 6 in this series; source in `papers/range/`).

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Citations are rendered from `results/s0_baseline/references.tsv`, numbered in order of first appearance.

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